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# Hierarchical semantic fusion in hyperbolic space for next POI recommendation

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ABSTRACT

Next Point-of-Interest (POI) recommendation aims to suggest POIs that users are likely to visit next based on their recent check-ins. Recent studies suggest that POI-POI transitions and user-POI interactions exhibit hierarchical patterns that can be modeled in hyperbolic space, whose exponential volume growth enables compact and expressive representations. However, existing hyperbolic methods insufficiently explore rich contextual semantics, such as category and region information, and thus fail to leverage these signals when modeling user mobility patterns. Besides, empirical analysis implies that check-in frequencies across regions and categories exhibit hierarchical distribution patterns, while both first-order and high-order POI transitions, along with their associated region and category transitions, also imply hierarchical patterns. We hypothesize that effectively exploiting these hierarchical semantics is essential for capturing user mobility patterns and enabling better recommendations. Thus, we propose H2POI, a unified hyperbolic framework that systematically fuses diverse hierarchical semantics through geometry-aware operations for next POI task. H2POI first performs global perspective fusion by integrating first-order POI-POI transitions and user interactions with contextual region and category semantics, thereby constructing expressive entity representations. Then, it proceeds from a trajectory perspective, incorporating region and category semantics to enhance the modeling of check-ins and to construct two complementary trajectory representations that comprehensively capture transition dependencies. Finally, it further incorporates region and category predictions as auxiliary supervision during training, complementing semantic fusion and strengthening next POI recommendation. Experiments on three real-world datasets show that H2POI consistently outperforms state-of-the-art baselines, demonstrating the benefit of jointly modeling hierarchical semantics for next POI recommendation.

## 1. Introduction

Recently, location-based social networks (LBSNs), such as Meituan and Foursquare, have gained significant popularity. Millions of users on various LBSNs share their locations with others by checking in at Points-of-Interest (POIs), leading to the increasing availability of check-in data. These data have attracted significant research attention, making POI recommendation (Sánchez & Bellogín, 2022) a widely studied topic that aims to understand complex user mobility patterns, thereby enhancing user experiences and supporting location-based services. Among various POI recommendation tasks, next POI recommendation (Cheng et al., 2013; Li et al., 2026a; Liu et al., 2016; Zhou et al., 2024) has been extensively studied, aiming to suggest POIs for a user to visit next, given their recent check-in trajectory. Accurate next POI recommendations are crucial for improving user satisfaction and the operational efficiency of LBSNs.

Next POI recommendation has been widely studied. Compared with traditional studies, methods (Wu et al., 2022; Yang et al., 2020) that leverage deep learning techniques offer significant advantages for mining diverse patterns in check-in data to model user mobility trends. Numerous studies (Chen et al., 2025b; Li et al., 2024; Yan et al., 2023; Yang et al., 2022c) have demonstrated that integrating graph

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learning with trajectory modeling can lead to remarkable improvements in next POI recommendations. Graph learning techniques excel at capturing the global structural characteristics of users and POIs, whereas trajectory modeling methods are particularly effective in modeling the sequential transition patterns of users across POIs over time. Despite their success, most existing methods operate in Euclidean space, limiting their ability to capture complex hierarchical patterns inherent in check-ins. In particular, user and POI frequencies often exhibit power-law distributions (Feng et al., 2020; Huang et al., 2024b), revealing hierarchical patterns in user mobility data. However, such patterns are difficult to efficiently represent in Euclidean space due to its polynomial volume growth (Peng et al., 2022), which limits the ability of Euclidean methods to capture the hierarchical characteristics of check-ins.

Instead, several recent studies (Feng et al., 2020; Huang et al., 2026; Liu et al., 2025; Qiao et al., 2025) have leveraged hyperbolic geometry for next POI recommendation. Owing to its negative curvature and exponential volume growth (Nickel & Kiela, 2017; Peng et al., 2022), hyperbolic space can effectively capture hierarchical patterns in user mobility data (Feng et al., 2020). This enables more structured entity representations, improving POI recommendation over equivalent Euclidean designs. Despite these advances, existing methods primarily focus on users and POIs themselves, with shallow exploration of rich contextual semantics, including POI categories and regional information. For instance, HME (Feng et al., 2020) mainly captures hierarchical user-POI relations and first-order POI-POI transition frequencies, while incorporating category and region as auxiliary signals. HVGAE (Liu et al., 2025) enhances structural representation quality by employing a hyperbolic variational graph autoencoder to reconstruct user-POI interactions. Huang et al. (2026) only model the hierarchical relationships between POIs and categories in hyperbolic space. Meanwhile, HMST (Qiao et al., 2025) introduces hyperbolic rotations to separately characterize POI transitions together with their induced category and region transitions. However, these methods either consider only a subset of semantics or integrate them in a loosely coupled manner, lacking a unified hyperbolic framework to systematically organize diverse semantics.

POI category and region information play a crucial role in next POI recommendation. Previous studies (Jia et al., 2024; Lim et al., 2022; Zang et al., 2022) have shown that fine-grained modeling of such contextual semantics can effectively mitigate data sparsity and thus enhance recommendation performance. This suggests that hyperbolic methods likewise need to account for these semantic characteristics in a fine-grained manner. Moreover, different semantic signals exhibit distinct hierarchical characteristics that existing hyperbolic POI methods have not sufficiently exploited. Especially, our empirical analysis on a real-world dataset (Section 3.2) shows that user check-ins are unevenly distributed across regions at different spatial levels, while POIs are also highly unevenly distributed across categories. Moreover, region and category transitions at different orders exhibit power-law patterns similar to those observed for POI transitions in previous work. Overall, region and category information characterizes where and what types of POIs users tend to visit, while exhibiting clear hierarchical characteristics in check-ins. Considering the importance of these contextual semantics and their intrinsic hierarchical structure, the systematic integration of rich and diverse semantics within hyperbolic space to model user mobility patterns remains largely unexplored.

In this work, we investigate how to integrate hierarchical semantics within hyperbolic space to characterize user mobility patterns. We leverage hyperbolic geometry to comprehensively fuse the rich semantics embedded in check-ins, including user-POI interactions, POI-POI transitions, POI categories, regional information, and temporal features. The hierarchical patterns observed across diverse semantics indicate that they could be effectively represented through the exponential growth property of hyperbolic space (Peng et al., 2022). Compared with modeling user and POI relations alone, incorporating hierarchical region and category semantics provides two key benefits. First, from a global perspective, hierarchical region and category semantics provide additional structural signals for characterizing POIs, leading to more informative POI representations and, subsequently, better characterization of user preferences. Second, from a trajectory perspective, incorporating region and category semantics into check-ins provides complementary spatial and categorical views of sequential transitions, thereby enabling more comprehensive modeling of user mobility patterns. However, jointly modeling heterogeneous semantics within a unified hyperbolic framework remains non-trivial. The central challenge lies in designing an effective fusion mechanism that integrates heterogeneous semantic components to model user mobility patterns.

Our solution is **H2POI**, a novel hyperbolic hierarchical semantic fusion framework for next POI recommendation. Inspired by recent advances in Euclidean geometry that unify graph learning and trajectory modeling, H2POI systematically integrates various semantics by jointly modeling global structural relations and sequential mobility dynamics. Specifically, H2POI consists of three main components. First, a *Global Semantic Fusion* module integrates heterogeneous POI relations and user-POI interactions in hyperbolic space to learn structurally expressive POI and user representations. Second, a *Trajectory Semantic Fusion* module disentangles and models categorical and spatial semantics from dual perspectives to construct two types of trajectory representations. This enables semantically differentiated and structurally consistent modeling of sequential mobility patterns. Finally, a *Multi-task Prediction* module predicts the next POI, category, and region based on the two types of trajectory representations. It incorporates category and region prediction as auxiliary objectives, thereby enhancing semantic integration and improving generalization in next POI recommendation.

Our main contributions can be summarized as follows:

- Through empirical data analysis, we reveal distinct hierarchical characteristics in region and category semantics beyond the user-POI relations mainly explored by existing hyperbolic POI methods. These findings motivate the joint exploitation of diverse hierarchical semantics in hyperbolic space to better characterize user mobility patterns.
- We propose a novel hyperbolic model, H2POI, that fuses multiple semantics that exhibit underlying hierarchical patterns within a unified hyperbolic framework for next POI recommendation. Specifically, H2POI integrates these semantics in check-in data at both global and trajectory levels and employs multi-task learning with additional semantic supervision to improve next POI recommendation performance.
- We conduct extensive experiments on three real-world check-in datasets. The empirical results demonstrate that H2POI significantly outperforms various existing baselines, demonstrating its effectiveness. The source code and datasets are available at https://github.com/PlaymakerQ/H2POI.

## 2. Related work

### 2.1. Next POI recommendation

The next POI recommendation task has attracted extensive research interest. Early studies focused on capturing users' static preferences for POIs through methods such as matrix factorization (Rendle et al., 2010; Ye et al., 2011) and embedding learning (Feng et al., 2015). In recent years, methods (Cui et al., 2026; Liu et al., 2016; Qi et al., 2024; Sun et al., 2020; Zhao et al., 2022) that use recurrent neural networks to model sequential dependencies have improved prediction accuracy by capturing dynamic characteristics. Subsequently, the significant success of transformers (Vaswani et al., 2017) in sequence tasks has led to their widespread adoption in next POI recommendation (Duan et al., 2023; Li et al., 2026a, 2025; Luo et al., 2021; Wang et al., 2022a), demonstrating improved trajectory modeling performance. Besides, graph learning

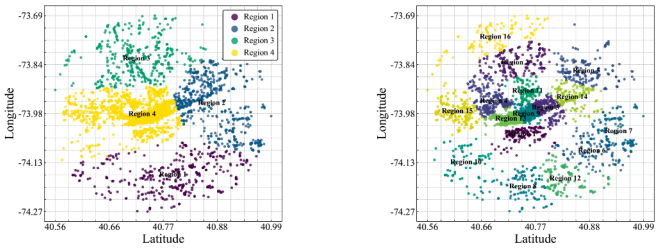
methods have been widely used to capture complex relations in check-in data. For instance, some studies mainly model relations among POIs to enhance POI representations (Qin et al., 2023; Wang et al., 2022b, 2023b), while others further incorporate richer contextual information to characterize user mobility (Lei et al., 2025; Mu et al., 2025; Rao et al., 2025).

Very recently, methods (Chen et al., 2025b; Li et al., 2024; Ma et al., 2024; Xu et al., 2024; Yan et al., 2023; Yang et al., 2022c; Zhuang et al., 2024) based on the framework combining graph learning and trajectory modeling have achieved state-of-the-art performance on next POI recommendation. For example, to capture collaborative signals from other users, GETNext (Yang et al., 2022c) constructs a trajectory flow graph and designs a graph-enhanced transformer model for predicting a user's future mobility. MCN4Rec (Li et al., 2024) effectively extracts multi-semantic correlations among various factors in check-in data through heterogeneous contrastive representation learning combined with transformer-based sequence modeling. Though effective, all these methods are based on Euclidean space, which limits their ability to fully capture the complex non-Euclidean nature of the data, thereby restricting their performance in the next POI tasks.

Check-in data contain rich hierarchical semantics that characterize user mobility (Chen et al., 2025a; Feng et al., 2020; Zang et al., 2022). Some studies (Jia et al., 2024; Lim et al., 2022; Wang et al., 2024b; Zang et al., 2022) explicitly model hierarchical information in check-in data, especially POI categories, and incorporate them into POI representations or trajectory modeling. Recent studies in other recommendation tasks have also explored diverse hierarchical information (He et al., 2025, 2026; Li et al., 2026b; Tang et al., 2026), but are not specifically designed for the characteristics of check-in data. More recently, several hyperbolic methods (Feng et al., 2020; Liu et al., 2025; Qiao et al., 2025) have mainly exploited the hierarchical structures of POI-POI and user-POI relations, with limited integration of diverse hierarchical semantics. Different from these studies, we systematically exploit hierarchical structures across diverse semantics and fuse them within a unified hyperbolic framework.

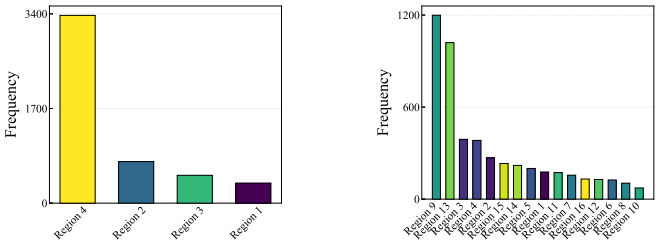
### 2.2. Hyperbolic recommender system

Hyperbolic geometry has demonstrated strong capability in modeling hierarchical structures (Chen et al., 2025c; Du et al., 2024). To capture implicit or explicit hierarchical patterns, hyperbolic techniques have been widely applied in recommendation tasks (Du et al., 2022; Sun et al., 2021). Recent developments include knowledge-aware recommendation (Chen et al., 2022b), social recommendation (Yang et al., 2024), session-based recommendation (Guo et al., 2023), and news recommendation (Wang et al., 2023a). For example, collaborative filtering techniques combined with hyperbolic space (Huang et al., 2024a; Li et al., 2023; Qin et al., 2025; Yang et al., 2022a,b) outperform conventional collaborative filtering methods. Yang et al. (2024) developed the HGSR framework to jointly extract user-user and user-item relations in hyperbolic space, enhancing the modeling of social factors. Besides, hyperbolic geometry has also been widely applied to other fields, such as knowledge graph completion (Chami et al., 2020; Wang et al., 2024a) and information diffusion (Qiao et al., 2023, 2026; Qiu et al., 2024). However, existing hyperbolic recommenders are ineffective for next POI recommendation, as they fail to capture diverse semantics or model sequential transitions in check-ins. Moreover, only a few studies (Feng et al., 2020; Huang et al., 2026; Liu et al., 2025; Qiao et al., 2025) have explored hyperbolic methods for next POI recommendation. However, these approaches focus on partially or separately modeling different semantic relations, which might prevent them from adequately exploiting the properties of hyperbolic geometry to reflect complex mobility patterns. Observing that multiple semantic signals exhibit inherent hierarchical properties, we propose to leverage a unified hyperbolic space to systematically fuse diverse hierarchical semantics for next POI recommendation.




(a) Level 1 (b) Level 2

**Fig. 1.** Illustration of two-level region partitioning on the NYC dataset.




(a) Level 1 (b) Level 2

**Fig. 2.** Distributions of check-in frequencies across different region splits on the NYC dataset.

## 3. Preliminary

### 3.1. Problem definition

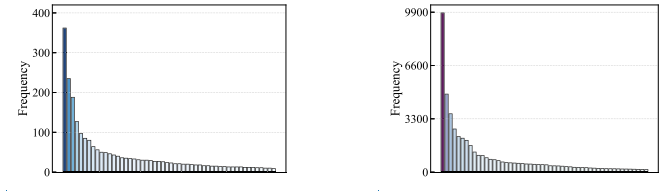
Let $\mathcal{U}$ and $\mathcal{L}$ denote the user set and the POI set, with $N_u = |\mathcal{U}|$ and $N_l = |\mathcal{L}|$ representing the number of users and POIs, respectively. Each POI $l \in \mathcal{L}$ has a category $c$, reflecting its categorical semantics (e.g., functions and attributes). Based on its geographical coordinates, we can obtain the POI's region $g$, which indicates its spatial semantics. We divide the whole area into $K$ levels using a clustering method to explicitly construct multi-level spatial semantics, while the uneven POI distributions across regions at each level exhibit implicit hierarchical characteristics. The clustering results at all levels are then used to assign region labels to POIs. The details of the multi-level region clustering will be introduced later. Let $\mathcal{C}$ denote the category set and $\mathcal{G}$ the region set, with $N_c = |\mathcal{C}|$ and $N_g = |\mathcal{G}|$.

A check-in $q = (u, l, t, c, g)$ denotes that the user $u$ visits the POI $l$ at time $t$, while $c$ and $g$ specify the categorical and spatial semantics of $l$, respectively. In this work, we represent $g$ using the region labels of $l$ across multiple spatial levels. A trajectory $s^u = \{q_1, q_2, \cdots, q_m\}$ contains a chronological list of check-ins of user $u$, where $m$ denotes the length and $q_i$ indicates the $i$-th check-in the trajectory.

**Next POI Recommendation Problem**: Given a set of historical check-ins and a recent trajectory $s^u$ of user $u$, the purpose of the next POI recommendation is to recommend a set of POIs that the user $u$ would visit subsequently.

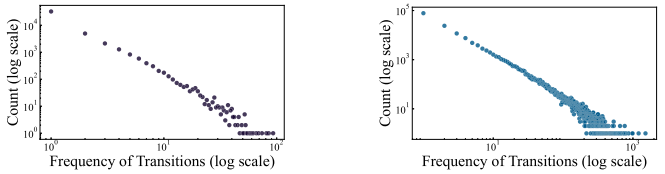
### 3.2. Data observation and analysis

Here, we conduct data analyses to reveal intrinsic characteristics of a check-in dataset NYC. Similar patterns can also be observed in other datasets used in this work (Section 5), including TKY and CA. A work, HME (Feng et al., 2020), has revealed power-law patterns in user and POI frequencies, providing initial evidence of hierarchical characteristics in user mobility data. However, these observations capture only a limited aspect of the hierarchical patterns in check-in data. We further provide comprehensive analyses of patterns across regions, categories, and both first- and high-order sequential transition relations.



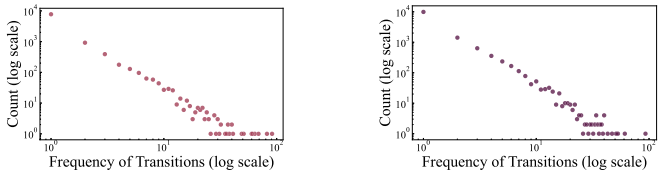

(a) POI frequency (b) Check-in frequency

**Fig. 3.** Distributions of POI and check-in frequencies across categories on the NYC dataset.




(a) First-order POI-POI (b) High-order POI-POI

**Fig. 4.** Illustration of inherent hierarchical patterns in first-order and high-order POI transition relations on the NYC dataset.




(a) Region-Region (b) Category-Category

**Fig. 5.** Illustration of inherent hierarchical patterns in high-order region and category transition relations on the NYC dataset.

First, we examine the spatial features of check-in data, which exhibit explicit and implicit hierarchical structures. We generate region labels for POIs by partitioning the whole area at multiple spatial granularities. Specifically, we use K-means (Hartigan & Wong, 1979) for multi-level region partitioning. Given a base $a$, the area is independently divided into $a^k$ clusters at level $1 \le k \le K$, and each POI is assigned a region label $g^k$ at each level. These levels represent different spatial granularities and are not required to form a nested tree structure. An illustration of the method with $a = 4$ and $K = 2$ is shown in Fig. 1. Compared with HME (Feng et al., 2020), which divides an area into fixed rectangular blocks at each level, the inherent spatial coherence between locations can be better preserved. Fig. 2 shows that users' check-ins are unevenly distributed at each hierarchical level across regions, which may imply latent hierarchical structures (Yang et al., 2022a).

Second, similar patterns are observed in the distributions of POIs and check-ins across POI categories, as shown in Fig. 3. The distribution of POIs across categories is inherently uneven. User preferences for specific POIs further increase the unevenness of the check-in distribution across categories. This disparity in POI and check-in frequencies highlights the varying popularity and user interest associated with POI categories. The heavily skewed distributions may indicate latent hierarchical structures, which can be effectively modeled in hyperbolic space (Nickel & Kiela, 2018; Nickel & Kiela, 2017; Yang et al., 2022a).

Moreover, we also observe implicit hierarchical patterns in various sequential transition relations. For simplicity, let $a \to b \to c$ represent a user's check-in trajectory. Here, $a \to b$ and $b \to c$ are first-order transitions, while $a \to c$ represents a high-order transition. Analyzing all trajectories, we found that both first-order and high-order POI-POI transition relations follow clear power-law distributions, as shown in Fig. 4. Region and category transitions accompanying POI transitions also exhibit similar characteristics. We present the high-order region-region and category-category transition relations in Fig. 5. The observed power-law distributions may reflect latent hierarchical structures in these transitions, yet such imbalanced patterns have often been overlooked in previous studies.

As discussed in previous sections, existing hyperbolic studies (Feng et al., 2020; Huang et al., 2026; Liu et al., 2025; Qiao et al., 2025) either partially capture or overlook the hierarchical patterns implied by these data characteristics. As a result, the capability of hyperbolic geometry to capture rich hierarchical semantics has not been properly exploited, and its potential benefits for next POI recommendation remain underutilized. To bridge this gap, we explore modeling all these hierarchies using hyperbolic geometry to enhance next-POI recommendations.

### 3.3. Hyperbolic geometry

In this work, we adopt the Lorentz model for describing hyperbolic spaces. Specifically, we represent $\mathbb{L}_\gamma^d$ as a $d$-dimensional Lorentz model with a negative curvature $-1/\gamma$, where $\gamma > 0$ is the curvature parameter. The point set in the Lorentz model is defined as $\mathbb{L}_\gamma^d = \{\mathbf{x} \in \mathbb{R}^{d+1} | \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -\gamma\}$, where $\mathbf{x} = (x_0, x_1, \cdots, x_d)$ is $d+1$ dimensional vector with $x_0 = \sqrt{\gamma + \sum_{i=1}^d x_i^2} > 0$. Note that $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} = -x_0 \cdot y_0 + \sum_{i=1}^d x_i \cdot y_i$ denotes the Lorentzian scalar product. Define the origin of the space as $\mathbf{o} = \{\sqrt{\gamma}, 0, \cdots, 0\} \in \mathbb{L}_\gamma^d$. For any $\mathbf{x} \in \mathbb{L}_\gamma^d$, there is an $d$-dimensional vector space $\mathcal{T}_{\mathbf{x}}\mathbb{L}_\gamma^d$, named the tangent space, centered at $\mathbf{x}$. We use exponential map $\exp_{\mathbf{x}}^\gamma : \mathcal{T}_{\mathbf{x}}\mathbb{L}_\gamma^d \to \mathbb{L}_\gamma^d$ and logarithmic map $\log_{\mathbf{x}}^\gamma : \mathbb{L}_\gamma^d \to \mathcal{T}_{\mathbf{x}}\mathbb{L}_\gamma^d$ to achieve the mapping between tangent space and hyperbolic space (Chami et al., 2019). The squared Lorentzian distance between $\mathbf{x}, \mathbf{y} \in \mathbb{L}_\gamma^d$ is defined as:

$$\mathrm{d}_{\mathcal{L}}^2(\mathbf{x}, \mathbf{y}) = -2\gamma - 2\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}. \tag{1}$$

Besides, given a set of hyperbolic vectors $\{\mathbf{x}_1, \cdots, \mathbf{x}_n\}$, where $\mathbf{x}_i \in \mathbb{L}_\gamma^d$, their hyperbolic centroid $\mathbf{x}^{\text{cen}}$ is computed as:

$$\mathbf{x}^{\text{cen}} = \sqrt{\gamma} \frac{\sum_{i=1}^n \mathbf{x}_i}{\left| \left\| \sum_{i=1}^n \mathbf{x}_i \right\|_{\mathcal{L}} \right|}, \tag{2}$$

where $|\|\mathbf{x}\|_{\mathcal{L}}| = \sqrt{|\|\mathbf{x}\|_{\mathcal{L}}^2|}$, $\mathbf{x} \in \mathbb{L}_\gamma^d$, denotes the modulus of the Lorentzian norm (Law et al., 2019).

We omit $\gamma$ and fix its value at $\gamma = 1$. We use the origin $\mathbf{o}$ as a reference point to perform tangent space transformations $\exp_{\mathbf{o}}(\cdot)$ and $\log_{\mathbf{o}}(\cdot)$ between $\mathcal{T}_{\mathbf{o}}\mathbb{L}^d$ and $\mathbb{L}^d$. For more details on hyperbolic space, please refer to the survey (Peng et al., 2022).

## 4. Methodology

Observing that different semantic signals in check-in data exhibit distinct hierarchical characteristics, we propose a unified hyperbolic framework **H2POI** that effectively models and fuses these diverse hierarchical semantics to characterize user mobility patterns. As illustrated in Fig. 6, H2POI models and integrates hierarchical semantics from both global and trajectory perspectives, thereby systematically fusing diverse semantics in hyperbolic space to infer users' future mobility trends.

### 4.1. Global semantic fusion

From a global perspective, different semantic signals provide complementary information for characterizing POIs through their contextual semantics and user preferences through interaction information. Existing Euclidean methods (Li et al., 2024; Wang et al., 2023b; Yang et al., 2022c) effectively exploit rich contextual information but are limited in capturing its hierarchical characteristics. Meanwhile, existing hyperbolic methods (Huang et al., 2026; Liu et al., 2025; Qiao et al., 2025) better capture hierarchical patterns but make limited use of diverse contextual semantics. To address these limitations, we design Global Semantic Fusion as a two-stage representation scheme that progressively

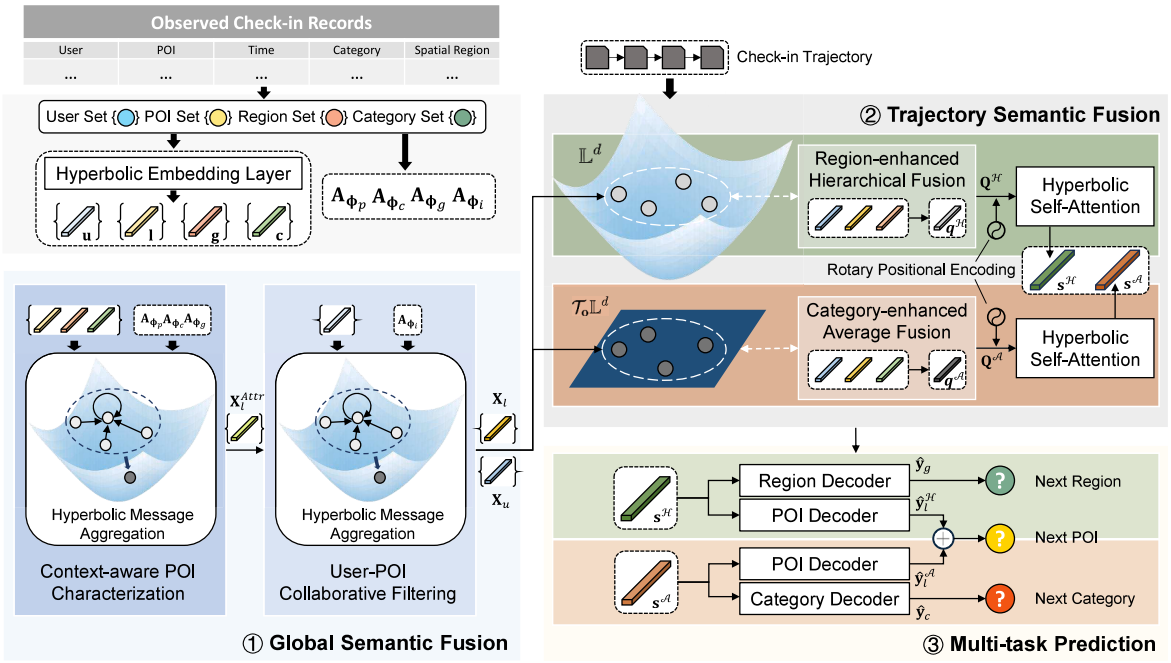



**Fig. 6.** The framework of H2POI. H2POI has three main components. First, *Global Semantic Fusion* integrates various POI relations and user-POI interactions from a global perspective to construct expressive hyperbolic representations for POIs and users. Second, *Trajectory Semantic Fusion* operates from a trajectory perspective by employing distinct fusion mechanisms to generate two types of contextually enhanced trajectory representations. Third, *Multi-Task Prediction* employs multiple decoders to predict the next POI, region, and category, respectively.

integrates first-order POI transition relations, POI-associated region and category information, and user-POI interactions in hyperbolic space.

Specifically, the two stages organize these semantics according to their roles in mobility characterization. In the first stage, we jointly model first-order POI transition relations and POI-associated region and category semantics to characterize location features. In the second stage, we integrate the resulting semantically enhanced POI representations with user-POI interactions to capture user preferences. In this way, we progressively incorporate contextual POI semantics into user-POI representations, providing semantically enriched representations for subsequent trajectory modeling.

To implement this two-stage semantic fusion, we construct three POI semantic relations from observed check-ins: the POI transition relation $\Phi_p$, categorical relation $\Phi_c$, and spatial relation $\Phi_g$, together with the user-POI interaction relation $\Phi_i$, as illustrated in Fig. 6. We initialize user, POI, category, and region embeddings in $\mathbb{L}^d$ using hyperbolic Gaussian sampling following Sun et al. (2021), Yang et al. (2022b). Both stages are then implemented through a common hyperbolic message aggregation operation.

**Hyperbolic Message Aggregation**. Following Yang et al. (2022b), for a given node $i$ with an initial embedding $\mathbf{x}_i$, the $\mathcal{K}$-layer hyperbolic neighborhood aggregation is performed by combining the representations of its neighbors from the preceding aggregation step. The $l$-th layer aggregation is calculated as:

$$\mathbf{z}_i^{\mathcal{T},l+1} = \mathbf{z}_i^{\mathcal{T},l} + \sum_{j\in\mathcal{N}_i} \frac{1}{|\mathcal{N}_i|}\mathbf{z}_j^{\mathcal{T},l}, \tag{3}$$

where $\mathcal{N}_i$ indicates the one-hop neighbors of $i$, $\mathbf{z}^{\mathcal{T},l} \in \mathcal{T}_{\mathbf{o}}\mathbb{L}^d$, and $\mathbf{z}_i^{\mathcal{T},0} = \log_{\mathbf{o}}(\mathbf{x}_i)$. The final aggregated representation is:

$$\mathbf{z}_i = \exp_{\mathbf{o}}\left(\sum_{l=1}^{\mathcal{K}} \mathbf{z}_i^{\mathcal{T},l}\right), \tag{4}$$

where $\mathcal{K}$ denotes the number of hyperbolic aggregation layers, and $\mathbf{z}_i$ is the final aggregated hyperbolic representation of node $i$. For convenience, we use $\mathbf{Z} = \mathrm{HypMess}(\mathbf{X}^0, \mathbf{A})$ to represent the aggregation process over the whole graph with $N$ nodes, where $\mathbf{X}^0 \in \mathbb{L}^{N\times d}$ is the initial embedding matrix, $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the adjacency matrix, and $\mathbf{Z} \in \mathbb{L}^{N\times d}$ denotes the aggregated hyperbolic representations at the final layer.

#### 4.1.1. Context-aware POI characterization

The first stage leverages contextual semantics, including region and category information, to construct POI representations for distinctive location characterization.

Let $\mathcal{R} = \{\Phi_p, \Phi_c, \Phi_g\}$. For a given relation $\Phi_r \in \mathcal{R}$, the related adjacency matrix is $\mathbf{A}_{\Phi_r} \in \mathbb{R}^{N_l\times N_l}$. We represent the message aggregation of POI representations for relation $\Phi_r$ as:

$$\mathbf{X}_l^{\Phi_r} = \mathrm{HypMess}(\mathbf{X}_l^0, \mathbf{A}_{\Phi_r}), \tag{5}$$

where $\mathbf{X}_l^0 = \{\mathbf{l}_1, \cdots, \mathbf{l}_{N_l}\} \in \mathbb{L}^{N_l\times d}$ is the initial POI embedding matrix. Then, we measure the importance of each relation using learnable weights. That is, we fuse POI representations with different POI relations through an attention mechanism:

$$\mathbf{X}_l^{Attr} = \sum_{\Phi_r\in\mathcal{R}} \beta_{\Phi_r} \cdot \mathbf{X}_l^{\Phi_r}, \tag{6}$$

where $\beta_{\Phi_r}$ is the attention score of relation $\Phi_r$. Inspired by Wang et al. (2021), $\beta_{\Phi_r}$ is calculated as:

$$\beta_{\Phi_r} = \frac{w_{\Phi_r}}{\sum_{\Phi_i\in\mathcal{R}} w_{\Phi_i}}, \tag{7}$$

$$w_{\Phi_r} = \frac{1}{N_l}\sum_{l_i\in\mathcal{L}} \mathbf{a}_{\Phi_r}^{\top} \cdot \tanh\left(\mathbf{W}_{\Phi_r}(\log_{\mathbf{o}}(\mathbf{x}_{l_i}^{\Phi_r}) \| \log_{\mathbf{o}}(\mathbf{h}_{l_i}^{\Phi_r})) + \mathbf{b}_{\Phi_r}\right), \tag{8}$$

where $\mathbf{a}_{\Phi_r} \in \mathbb{R}^{2d\times 1}$ is the attention vector for relation $\Phi_r$, $\|$ denotes the concatenation operation, $\mathbf{W}_{\Phi_r} \in \mathbb{R}^{2d\times 2d}$ and $\mathbf{b}_{\Phi_r} \in \mathbb{R}^{2d\times 1}$ are learnable parameters for relation $\Phi_r$, $\mathbf{x}_{l_i}^{\Phi_r} \in \mathbf{X}_l^{\Phi_r}$, and $\mathbf{h}_{l_i}^{\Phi_r}$ denotes the related initial embedding for relation $\Phi_r$. Here, $\mathbf{h}_{l_i}^{\Phi_r}$ equals $\mathbf{l}_i$, $\mathbf{c}_{l_i}$, or $\mathbf{g}_{l_i}$ based on whether $\Phi_r$ equals $\Phi_p$, $\Phi_c$, or $\Phi_g$, respectively. $\mathbf{l}_i$, $\mathbf{c}_{l_i}$, and $\mathbf{g}_{l_i}$ denote the initial POI embedding, its associated category embedding, and its aggregated region embedding, respectively. Note that for $\Phi_g$, each POI

$l_i$ corresponds to multiple region labels and, thus, multiple region embeddings. We calculate $\mathbf{g}_{l_i}$ as the centroid of the region embeddings of $l_i$ across the $K$ levels using Eq. (2). Therefore, $\mathbf{X}_l^{Attr}$ denotes the updated POI representations that integrate hierarchical semantics from first-order transitions, POI-associated region and category relations.

#### 4.1.2. User-POI collaborative filtering

The second stage uses the enhanced POI representations to model user-POI interactions. This design enables more accurate characterization of user preferences and further refines POI representations through collaborative signals.

Similarly, preference collaborative filtering is conducted through hyperbolic message aggregation as:

$$\begin{bmatrix} \mathbf{X}_u \\ \mathbf{X}_l \end{bmatrix} = \mathrm{HypMess}(\begin{bmatrix} \mathbf{X}_u^0 \\ \mathbf{X}_l^{Attr} \end{bmatrix}, \mathbf{A}_{\Phi_i}), \tag{9}$$

where $\mathbf{X}_u^0 = \{\mathbf{u}_1, \cdots, \mathbf{u}_{N_u}\} \in \mathbb{L}^{N_u \times d}$ is the initial user embedding matrix, $\mathbf{X}_l^{Attr}$ is from Eq. (6) in POI relation fusion, and $\mathbf{A}_{\Phi_i}^{N_{ul} \times N_{ul}}$ is the adjacency matrix for $\Phi_i$, where $N_{ul} = N_u + N_l$. $\mathbf{X}_u$ and $\mathbf{X}_l$ represent learned user and POI hyperbolic representations, respectively.

Subsequently, we will use these hyperbolic representations for user trajectory modeling.

### 4.2. Trajectory semantic fusion

We further perform hierarchical semantic fusion at the trajectory level to capture long-term mobility dynamics. Existing hyperbolic methods, such as Qiao et al. (2025), integrate different contextual semantics uniformly. However, this strategy may not exploit the distinct semantic characteristics observed in our data analysis (Section 3.2). To address this limitation, we propose a complementary trajectory modeling scheme that separately exploits the distinct characteristics of region and category semantics in hyperbolic space.

Given a user trajectory, we use the last check-in as the query in hyperbolic self-attention to model high-order sequential dependencies among observed check-ins. The core idea is to construct two types of check-in representations based on category semantics and region semantics, respectively, thereby forming two semantically complementary trajectory representations that preserve diverse semantics. Rather than compressing multiple semantics into a single trajectory representation, this design separates distinct semantic perspectives to prevent potential information entanglement. Under hyperbolic geometry, this design preserves the hierarchical structure of the data while enabling effective semantic disentanglement, thereby providing a more comprehensive characterization of complex temporal dependencies in user mobility.

Specifically, the module first constructs two types of check-in representations through different hyperbolic operations. It then incorporates temporal information via rotary positional encoding. Finally, separate hyperbolic self-attention modules are applied to obtain the final trajectory representations. Given a user trajectory $s^u = \{q_1, q_2, \cdots, q_m\}$, where each check-in is given by $q_i = (u, l_i, t_i, c_{l_i}, g_{l_i})$, we next describe how the two types of trajectory representations are constructed for $s^u$.

#### 4.2.1. Dual-perspective check-in representation

We use two hyperbolic operations, hierarchical fusion and average fusion, to construct two types of check-in representations, which enable effective preservation and differentiation of the contextual semantics of check-ins.

**Region-enhanced Hierarchical Fusion.** Given the explicit hierarchical structure of regions and their alignment with the hierarchical characteristics of POIs, hyperbolic centroid computation is employed to fuse user, POI, and region representations, thereby preserving structural consistency in hyperbolic space. The user and POI representations are fused with region embeddings across multiple layers via hyperbolic centroid aggregation to generate a region-aware check-in representation $\mathbf{q}_i^{\mathcal{H}}$ for $q_i$ as follows:

$$\mathbf{q}_i^{\mathcal{H}} = \sqrt{\gamma} \cdot \frac{\mathbf{x}_u + \mathbf{x}_{l_i} + \sum_{k=1}^{K} \mathbf{g}_{l_i}^k}{\left| ||\mathbf{x}_u + \mathbf{x}_{l_i} + \sum_{k=1}^{K} \mathbf{g}_{l_i}^k||_{\mathcal{L}} \right|}, \tag{10}$$

where $\mathbf{x}_u$ from $\mathbf{X}_u$ is the user representation, $\mathbf{x}_{l_i}$ from $\mathbf{X}_l$ is the POI representation. For the given POI $l_i$, we obtain all corresponding hyperbolic region embeddings $(\mathbf{g}_{l_i}^1, \cdots, \mathbf{g}_{l_i}^K)$ based on the $K$-level region label $g_{l_i} = (g_{l_i}^1, \cdots, g_{l_i}^K)$ of $l_i$.

**Category-enhanced Average Fusion.** In contrast to the hierarchical fusion employed for region semantics, category semantics are integrated through tangent-space averaging to preserve their distinct structural properties. This design maintains semantic diversity across perspectives and facilitates complementary trajectory representations. To this end, we fuse the user and POI representations with the category embedding in the tangent space to construct a category-aware check-in representation $\mathbf{q}_i^{\mathcal{A}} \in \mathbb{L}^d$ for $q_i \in s^u$:

$$\mathbf{q}_i^{\mathcal{A}} = \exp_{\mathbf{o}}\Big(\mathrm{Mean}\big(\log_{\mathbf{o}}(\mathbf{x}_u) + \log_{\mathbf{o}}(\mathbf{x}_{l_i}) + \log_{\mathbf{o}}(\mathbf{c}_{l_i})\big)\Big), \tag{11}$$

where $\mathbf{x}_u$ and $\mathbf{x}_{l_i}$ are user and POI representations, respectively. $\mathbf{c}_{l_i}$ is the hyperbolic category embedding for $l_i$, and $\mathrm{Mean}(\cdot)$ denotes the vector-wise average.

#### 4.2.2. Dual-perspective trajectory fusion

Based on the dual-perspective check-in representations, we incorporate temporal information through rotary positional encoding and then employ hyperbolic self-attention to fuse the check-ins into trajectory representations.

**Rotary Positional Encoding** The temporal information of check-ins within a trajectory is crucial for capturing movement patterns. However, temporal features are difficult to map into hyperbolic space. Therefore, inspired by Chami et al. (2020), Feng et al. (2024), hyperbolic rotations are employed to perform positional encoding that quantizes temporal features. Note that rotation operations can model complex features without affecting the hierarchy of representation (Chami et al., 2020).

In detail, we construct a learnable block-diagonal rotation matrix $\mathrm{Rot}_{t_i}$ for $t_i$ of $q_i$ and incorporate temporal information into the check-in representations through varying rotation angles. The temporal-rotated check-in representations are:

$$\mathbf{q}_i^{\mathcal{H},T} = \mathrm{Rot}_{t_i}^{\mathcal{H}} \circ \mathbf{q}_i^{\mathcal{H}}, \tag{12}$$

$$\mathbf{q}_i^{\mathcal{A},T} = \mathrm{Rot}_{t_i}^{\mathcal{A}} \circ \mathbf{q}_i^{\mathcal{A}}, \tag{13}$$

where $\mathrm{Rot}_{t_i}^{\mathcal{A}}$ and $\mathrm{Rot}_{t_i}^{\mathcal{H}}$ are two time-related trainable rotation matrices. $\mathrm{Rot} \circ \mathbf{e}$ means applying rotation operation Rot on $\mathbf{e}$.

After rotary positional encoding, two groups of check-in representations, $\mathbf{Q}_{1:m}^{\mathcal{H},T} = [\mathbf{q}^{\mathcal{H},T}1, \cdots, \mathbf{q}^{\mathcal{H},T}m]$ and $\mathbf{Q}_{1:m}^{\mathcal{A},T} = [\mathbf{q}^{\mathcal{A},T}1, \cdots, \mathbf{q}^{\mathcal{A},T}m]$ are constructed for the given $s^u$. Subsequently, we construct two types of trajectory representations using hyperbolic self-attention.

**Hyperbolic Self-Attention.** The core idea of hyperbolic attention is to perform a weighted average of group vectors within hyperbolic space. Inspired by Chen et al. (2022a), the hyperbolic self-attention for a group of check-in representations $\mathbf{Q}_{1:m} = [\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_m]$ is:

$$\mathbf{s} = \sqrt{\gamma} \cdot \frac{\sum_{i=1}^{m} a_i \mathbf{q}_i^{\mathbf{v}}}{\left| ||\sum_{i=1}^{m} a_i \mathbf{q}_i^{\mathbf{v}}||_{\mathcal{L}} \right|}, \tag{14}$$

$$a_i = \frac{\exp\left(\frac{-\mathrm{d}_{\mathcal{L}}^2(\mathbf{q}_m^{\mathbf{q}}, \mathbf{q}_i^{\mathbf{k}})}{\sqrt{d}}\right)}{\sum_{k=1}^{m} \exp\left(\frac{-\mathrm{d}_{\mathcal{L}}^2(\mathbf{q}_m^{\mathbf{q}}, \mathbf{q}_k^{\mathbf{k}})}{\sqrt{d}}\right)}, \tag{15}$$

where $\mathbf{q}_i \in \mathbb{L}^d$, $\mathbf{q}^{\mathbf{q}} = \mathrm{HL}^q(\mathbf{q})$, $\mathbf{q}^{\mathbf{k}} = \mathrm{HL}^k(\mathbf{q})$ and $\mathbf{q}^{\mathbf{v}} = \mathrm{HL}^v(\mathbf{q})$. Note that $\mathrm{HL}^q(\cdot)$, $\mathrm{HL}^k(\cdot)$, and $\mathrm{HL}^v(\cdot)$ are three independent Lorentzian linear layers (Chen et al., 2022a) that transform the input into query, key, and value representations in the Lorentz model. To simplify, let $\mathbf{s} =$

$\text{HypAtt}(\mathbf{Q}_{1:m})$ represent the hyperbolic self-attention operation on an embedding matrix.

Finally, we use hyperbolic self-attention to generate the trajectory representations $\mathbf{s}^{\mathcal{H}}$ and $\mathbf{s}^{\mathcal{A}}$ in $\mathbb{L}^d$ for $s^u$ from two semantic perspectives:

$$\mathbf{s}^{\mathcal{H}} = \text{HypAtt}_{\mathcal{H}}(\mathbf{Q}^{\mathcal{H},T}), \tag{16}$$

$$\mathbf{s}^{\mathcal{A}} = \text{HypAtt}_{\mathcal{A}}(\mathbf{Q}^{\mathcal{A},T}), \tag{17}$$

where the subscripts of $\text{HypAtt}(\cdot)$ indicate that we utilize two independent attention layers. Note that both $\mathbf{s}^{\mathcal{H}}$ and $\mathbf{s}^{\mathcal{A}}$ are trajectory representations of $s^u$, containing similar yet distinct semantic meanings. Their similarity lies in modeling the same trajectory to capture transition dependencies, whereas their difference arises from the distinct semantic emphasis of each representation, resulting from the different construction of check-in representations. This dual perspective allows us to explore and integrate diverse semantics, thereby enhancing the model's generalization and improving predictive performance. Next, we make the final predictions based on the trajectory representations.

### 4.3. Multi-task prediction

The distinct characteristics of region and category semantics allow their corresponding supervision signals to provide complementary guidance for representation learning. By introducing region and category prediction as auxiliary tasks, such semantic-specific supervision further promotes the comprehensive and effective fusion of diverse semantics in hyperbolic space. This enables more expressive trajectory representations to capture user mobility patterns for next POI recommendation.

Specifically, we use multiple decoders on $\mathbf{s}^{\mathcal{A}}$ and $\mathbf{s}^{\mathcal{H}}$ to simultaneously predict the next POI, the next region, and the next category. The recommendation score $\hat{\mathbf{y}}_l \in \mathbb{R}^{1\times N_l}$ for the next POI is:

$$\hat{\mathbf{y}}_l^{\mathcal{H}} = \left(\log_{\mathbf{o}}(\mathbf{s}^{\mathcal{H}})\right)^{\top}\mathbf{W}_{\mathcal{H}} + \mathbf{b}_{\mathcal{H}}^{\top}, \tag{18}$$

$$\hat{\mathbf{y}}_l^{\mathcal{A}} = \left(\log_{\mathbf{o}}(\mathbf{s}^{\mathcal{A}})\right)^{\top}\mathbf{W}_{\mathcal{A}} + \mathbf{b}_{\mathcal{A}}^{\top}, \tag{19}$$

$$\hat{\mathbf{y}}_l = \alpha \cdot \hat{\mathbf{y}}_l^{\mathcal{A}} + (1-\alpha)\cdot \hat{\mathbf{y}}_l^{\mathcal{H}}. \tag{20}$$

where $\alpha$ is the score weight. Meanwhile, the recommendation scores for the next category and next region are $\hat{\mathbf{y}}_g \in \mathbb{R}^{1\times N_g^K}$ and $\hat{\mathbf{y}}_c \in \mathbb{R}^{1\times N_c}$:

$$\hat{\mathbf{y}}_g = \left(\log_{\mathbf{o}}(\mathbf{s}^{\mathcal{H}})\right)^{\top}\mathbf{W}_g + \mathbf{b}_g^{\top}. \tag{21}$$

$$\hat{\mathbf{y}}_c = \left(\log_{\mathbf{o}}(\mathbf{s}^{\mathcal{A}})\right)^{\top}\mathbf{W}_c + \mathbf{b}_c^{\top}. \tag{22}$$

$\mathbf{W}_{\mathcal{H}}, \mathbf{W}_{\mathcal{A}} \in \mathbb{R}^{d\times N_l}$, $\mathbf{W}_g \in \mathbb{R}^{d\times N_g^K}$, and $\mathbf{W}_c \in \mathbb{R}^{d\times N_c}$ are trainable weights. $\mathbf{b}_{\mathcal{H}}, \mathbf{b}_{\mathcal{A}} \in \mathbb{R}^{N_l\times 1}$, $\mathbf{b}_g \in \mathbb{R}^{N_g^K\times 1}$, $\mathbf{b}_c \in \mathbb{R}^{N_c\times 1}$ are prediction score compensations. For region labels, to enhance efficiency and accuracy, we use the labels at the level $K$ as the prediction labels. $N_g^K$ denotes the number of region labels at level $K$.

To train the model, we use cross-entropy as the loss function for all prediction results. The loss function $\mathcal{L}$ is:

$$\mathcal{L} = \mathcal{L}_l + \beta\cdot\left(\mathcal{L}_g + \mathcal{L}_c\right), \tag{23}$$

where $\mathcal{L}_l$, $\mathcal{L}_g$, and $\mathcal{L}_c$ are the losses for predicting the next POI, region, and category, respectively. $\beta$ is the loss weight.

### 4.4. Time complexity analysis

The time complexity of the proposed method is analyzed as follows. For Global Semantic Fusion, the complexity is $\mathcal{O}(|\mathcal{E}|d)$, where $|\mathcal{E}|$ is the number of edges in the heterogeneous graph and $d$ denotes the dimension. This is comparable to standard Euclidean GNNs, with only minor overhead from vectorized hyperbolic operations (e.g., $\exp_{\mathbf{o}}(\cdot)$ and $\log_{\mathbf{o}}(\cdot)$). In Trajectory Semantic Fusion, constructing two types of check-in representations and applying hyperbolic self-attention scale linearly with trajectory length $m$ and dimension $d$, resulting in $\mathcal{O}(md)$. Temporal rotations also incur minimal cost through efficient block-diagonal multiplications. The Multi-task Prediction module mainly involves matrix-vector multiplications for POI, region, and category predictions, with a total complexity of $\mathcal{O}(dN)$, where $N$ denotes the total number of prediction targets. Overall, the total time complexity of H2POI is linear with respect to the graph size and trajectory length, i.e., $\mathcal{O}(|\mathcal{E}|d + md + dN)$.

**Table 1**
Dataset statistics.

| | #User | #POI | #Cat | #Check-in | #Trajectory |
|---|---|---|---|---|---|
| NYC | 1048 | 4981 | 318 | 103,941 | 14,130 |
| TKY | 2282 | 7833 | 290 | 405,000 | 65,499 |
| CA | 3957 | 9690 | 296 | 238,369 | 45,123 |

## 5. Experiments

In this section, we validate the effectiveness of our model on three real-world datasets. We conduct various experiments to investigate the following questions: **RQ1**: Can H2POI perform effectively in the next POI prediction tasks? **RQ2**: How does each module of H2POI affect its performance? **RQ3**: How do region and category semantics influence next POI recommendation? **RQ4**: How efficient is H2POI in terms of training and inference time? **RQ5**: What information can be inferred when the parameters of H2POI change? **RQ6**: Can we intuitively illustrate captured hierarchical patterns?

### 5.1. Experimental setups

#### 5.1.1. Datasets

We use three common real-world datasets: FourSquare-NYC (Yang et al., 2015), FourSquare-TKY (Yang et al., 2015), and Gowalla-CA (Yuan et al., 2013). Similar to Yang et al. (2022c), we exclude users and POIs with fewer than 10 check-ins for all three datasets. Users' check-ins were split into trajectories with 24-hour intervals. Trajectories with only one check-in are removed. Each dataset is then divided into training, validation, and test sets in chronological order, with 80% for training, 10% for validation, and 10% for testing. The statistics of the three datasets after pre-processing are shown in Table 1.

#### 5.1.2. Baselines

We compare H2POI with the following representative or state-of-the-art methods to evaluate its effectiveness. The baselines can be categorized into two main classes: (1) Euclidean methods: **FPMC** (Rendle et al., 2010) is a classic method using the Markov chain to model user preferences. **LSTM** (Hochreiter & Schmidhuber, 1997), **PLSPL** (Wu et al., 2022), **Flashback** (Yang et al., 2020), and **CLSPRec** (Duan et al., 2023) are four methods possessing sequence modeling as their key components. **Graph-Flashback** (Rao et al., 2022), **GETNext** (Yang et al., 2022c), **AGRAN** (Wang et al., 2023b), **MCN4Rec** (Li et al., 2024), and **LoTNext** (Xu et al., 2024) are the next POI recommendation methods that integrate trajectory modeling and graph learning techniques. GETNext is a competitive method, while CLSPRec, AGRAN, MCN4Rec, and LoTNext are the four most recent baselines. (2) Hyperbolic methods: Given the limited exploration of hyperbolic geometry for next POI recommendation, we also adapt representative hyperbolic recommenders for a broader comparison. **HME** (Feng et al., 2020) is a hyperbolic method that relies solely on shallow network embeddings to model various relations equally. **HICF** (Yang et al., 2022a) and **HGSR** (Yang et al., 2024) represent two advanced hyperbolic recommendation methods. While they were not originally designed for POI recommendation, we adapt them for our task. Specifically, for HGSR, we establish user-user connections based on similar user preferences. **HMST** (Qiao et al., 2025) is a recent hyperbolic approach that models transitions under different semantics separately using hyperbolic rotations.

#### 5.1.3. Implementation details

All baselines are configured based on the original papers or their open-source implementations. For our method, we set the key hyper-

parameters as follows: We set $a = 4$ and $K = 4$ for multi-level region clustering across all datasets. We use the Adam optimizer with a learning rate of 0.003. The batch size is 64. The default curvature parameter is $\gamma = 1$ and the dimension is $d = 64$. The default number of layers for hyperbolic message aggregation is $\mathcal{K} = 2$. The hyperparameters, score weight $\alpha$ and loss weight $\beta$, are tuned for each dataset using grid search, with the search range $[0, 1]$ and step size 0.1. We use two common evaluation metrics in the next POI recommendation research, Top-$k$ Accuracy Rate (Acc@$k$) and Mean Reciprocal Rank (MRR), to measure the models' prediction precision and ranking quality. Our method is implemented in Python and PyTorch on an Ubuntu server with an Intel(R) i9-10900X 3.70GHz CPU and two NVIDIA GTX 4090 GPUs.

### 5.2. Overall performance

We begin by answering **RQ1**, where we compare H2POI with all baselines across three datasets to evaluate its effectiveness in the next POI recommendation tasks. The overall performance results on these datasets are summarized in Table 2. The metrics Acc@$k$ ($k = 1, 5, 10$) and MRR are reported for each dataset. We have several observations: (1) Among Euclidean baselines, methods combining sequence modeling with graph learning generally outperform other framework designs. Compared to the traditional method FPMC, sequential methods LSTM, PLSPL, Flashback, and CLSPRec achieve huge improvements, highlighting the significance of modeling check-in trajectories. Methods from Graph-Flashback to LoTNext enhance trajectory modeling by leveraging graph learning techniques to capture global relations, further improving recommendation accuracy in most cases. This indicates that such a combined framework is effective for the next POI recommendation. (2) H2POI consistently outperforms all Euclidean baselines by a significant margin. This result highlights the advantage of hyperbolic geometry in modeling user mobility. H2POI operates in hyperbolic space, which captures both implicit and explicit hierarchical patterns inherent in check-in data from both global and trajectory aspects. The exponential expansion property of hyperbolic geometry enables more compact and faithful representations of complex patterns, which results in a better characterization of user mobility trends and thus improved next POI prediction. (3) H2POI demonstrates clear advantages over four state-of-the-art hyperbolic recommendation methods. HICF and HGSR are not specifically tailored for POI recommendations and, therefore, fail to incorporate complex contextual semantics and spatial-temporal features in check-ins. Among hyperbolic POI methods, HME relies on hyperbolic embeddings for representation learning but does not model trajectories. HMST further models trajectories using single-perspective hyperbolic attention. In contrast, H2POI uses a hyperbolic heterogeneous graph neural network to model global relations and dual-perspective hyperbolic attention to model trajectories. This allows H2POI to better capture hierarchical semantics at both levels, improving performance over HME and HMST. Overall, H2POI consistently outperforms all baselines on three datasets, highlighting its superiority and effectiveness for the next POI recommendation.

### 5.3. Ablation study

We evaluate the effectiveness of each component of H2POI to answer **RQ2** through a comprehensive ablation study. Specifically, we implement nine ablation settings, with all other configurations using default settings except in the ablation scenarios. The ablation results are presented in Table 3, with the complete H2POI model serving as a reference. The ablation results for the multi-task prediction module are separately reported in Table 4.

From Table 3, we make the following observations. First, **w/o HYP** denotes implementing the whole H2POI in Euclidean space. The significant decline across all metrics suggests that hyperbolic space more effectively captures the rich hierarchical semantics in check-in data than its Euclidean counterpart. **w/o Region** and **w/o Category** represent variants where region or category semantics are completely removed. The performance drops to some extent in both cases, indicating that these contextual semantics provide useful signals for next POI recommendation. These results highlight the importance of hierarchical semantics and demonstrate the effectiveness of hyperbolic space for hierarchical semantic fusion in next POI recommendation.

Second, results (4)-(6) pertain to the global semantic fusion module. When we remove the entire module and retain only a hyperbolic embedding layer, as shown in **w/o GSF**, we observe a notable performance drop, particularly on the NYC and CA datasets. This result shows that, compared with using hyperbolic embeddings, the global semantic fusion module constructs more effective hyperbolic representations of users and POIs by fusing hierarchical semantics from global relations. Meanwhile, excluding either the context-aware POI characterization (**w/o CPC**) or the user-POI collaborative filtering (**w/o UPCF**) leads to a decline in performance across all cases, suggesting that both modules are complementary and essential for constructing expressive representations.

Third, results (7)-(9) investigate the effectiveness of the trajectory semantic fusion module. **w/o TSF** shows the results when the whole multi-semantic trajectory modeling module is removed, and the last check-in is used as the query to predict the next POI. This leads to a significant decline in prediction accuracy due to the inability to capture dynamic dependencies within trajectories. **w/o RHF** removes the region-enhanced hierarchical fusion, and **w/o CAF** removes the category-enhanced semantic average fusion. Consequently, we use a single fusion method, either Eq. (11) or Eq. (10), to integrate all semantics when constructing check-in representations. All metrics decline under both settings across all datasets. These results suggest that specific semantics can be reflected by various hyperbolic operations. Thus, our method outperforms others by leveraging a multi-semantic integration strategy that captures the complementary strengths of diverse semantics in hyperbolic space.

Table 4 presents the ablation results of the multi-task prediction module. We can observe that incorporating either region or category prediction consistently improves POI prediction performance across all datasets, while jointly incorporating both auxiliary tasks achieves the best performance. These results show that H2POI effectively leverages rich semantic supervision from region and category information to improve next POI recommendation.

In summary, the ablation study results indicate that the best performance is achieved when the model is fully intact.

### 5.4. Multi-task prediction analysis

For **RQ3**, we conduct experiments to investigate the impact of region and category predictions. We report POI, region, and category prediction results of H2POI across three datasets in Table 5, using Acc@1 and MRR as evaluation metrics. We observe that H2POI achieves higher prediction accuracy for region and category tasks than for POI prediction. The results indicate that, compared with sparse POIs, region and category information can be predicted more reliably. H2POI can effectively infer the category and region of the next potential location. The high region prediction accuracy indicates that coarse-grained spatial semantics provide a relatively reliable auxiliary signal for next POI prediction. In addition, these results suggest that H2POI's performance gains (as shown in Table 2) in the next POI recommendation may stem from its ability to leverage contextual semantics to highlight POIs that are more relevant to these attributes, thereby enabling more effective recommendations. The overall results validate that our model effectively exploits diverse semantics in hyperbolic space, thereby yielding strong predictive performance in practice.

**Table 2**
Overall performance comparison in Acc@k and MRR on three datasets. The best scores are in **boldface** and the second best scores are underlined. The percentage improvements over the best baseline scores are presented in the last row.

| | NYC | | | | TKY | | | | CA | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Acc@1 | Acc@5 | Acc@10 | MRR | Acc@1 | Acc@5 | Acc@10 | MRR | Acc@1 | Acc@5 | Acc@10 | MRR |
| FPMC | 0.1073 | 0.2758 | 0.3461 | 0.1845 | 0.0924 | 0.2379 | 0.3156 | 0.1550 | 0.0572 | 0.1403 | 0.1804 | 0.1019 |
| LSTM | 0.1399 | 0.3276 | 0.3895 | 0.2299 | 0.1585 | 0.3276 | 0.4087 | 0.2431 | 0.0824 | 0.2100 | 0.2641 | 0.1442 |
| PLSPL | 0.1918 | 0.4399 | 0.5323 | 0.2961 | 0.1991 | 0.3983 | 0.4804 | 0.2934 | 0.1052 | 0.2642 | 0.3343 | 0.1847 |
| Flashback | 0.1997 | 0.4849 | 0.5895 | 0.3305 | 0.1707 | 0.4254 | 0.5394 | 0.2870 | 0.1085 | 0.2595 | 0.3318 | 0.1800 |
| CLSPRec | 0.2179 | 0.4252 | 0.5018 | 0.3132 | 0.1750 | 0.3287 | 0.3721 | 0.2463 | 0.1192 | 0.2595 | 0.3432 | 0.1917 |
| Graph-Flashback | 0.2046 | 0.4931 | 0.5972 | 0.3339 | 0.1731 | 0.4273 | 0.5431 | 0.2905 | 0.1122 | 0.2626 | 0.3450 | 0.1827 |
| GETNext | 0.2274 | 0.4997 | 0.6017 | 0.3478 | 0.2178 | 0.4432 | 0.5383 | 0.3148 | 0.1231 | 0.3050 | 0.3853 | 0.2073 |
| AGRAN | 0.2148 | 0.4598 | 0.5508 | 0.3175 | 0.1878 | 0.4075 | 0.4967 | 0.2907 | 0.1141 | 0.2859 | 0.3879 | 0.1990 |
| MCN4Rec | 0.2282 | 0.4577 | 0.5368 | 0.3297 | 0.2239 | 0.4278 | 0.5052 | 0.3047 | 0.1199 | 0.2833 | 0.3589 | 0.1946 |
| LoTNext | 0.2152 | 0.5119 | 0.6113 | 0.3451 | 0.2016 | 0.4407 | 0.5441 | 0.3132 | 0.1364 | 0.3093 | 0.3929 | 0.2184 |
| HME | 0.1601 | 0.4120 | 0.5409 | 0.2746 | 0.1185 | 0.3538 | 0.4460 | 0.2241 | 0.1043 | 0.2788 | 0.3650 | 0.1869 |
| HICF | 0.1126 | 0.3447 | 0.4727 | 0.2161 | 0.0816 | 0.2597 | 0.3407 | 0.1677 | 0.0834 | 0.2534 | 0.3367 | 0.1675 |
| HGSR | 0.1724 | 0.4437 | 0.5512 | 0.2908 | 0.1262 | 0.3274 | 0.4542 | 0.2264 | 0.0950 | 0.2446 | 0.3214 | 0.1690 |
| HMST | 0.2188 | 0.5388 | 0.6233 | 0.3567 | 0.2018 | 0.4714 | 0.5633 | 0.3315 | 0.1429 | 0.3287 | 0.4200 | 0.2309 |
| H2POI | **0.2535** | **0.5519** | **0.6490** | **0.3830** | **0.2527** | **0.5084** | **0.6020** | **0.3645** | **0.1677** | **0.3720** | **0.4816** | **0.2597** |
| Improv. | 11.09% | 2.43% | 4.12% | 7.37% | 12.86% | 7.85% | 6.87% | 9.95% | 17.35% | 13.17% | 14.67% | 12.47% |

**Table 3**
The experimental results of the ablation study.

| | NYC | | TKY | | CA | |
|---|---|---|---|---|---|---|
| | Acc@1 | MRR | Acc@1 | MRR | Acc@1 | MRR |
| (1) w/o HYP | 0.2216 | 0.3563 | 0.2193 | 0.3377 | 0.1567 | 0.2377 |
| (2) w/o Region | 0.2367 | 0.3734 | 0.2338 | 0.3487 | 0.1443 | 0.2374 |
| (3) w/o Category | 0.2410 | 0.3713 | 0.2352 | 0.3510 | 0.1495 | 0.2387 |
| (4) w/o GSF | 0.1997 | 0.3172 | 0.2363 | 0.3343 | 0.1312 | 0.2182 |
| (5) w/o CPC | 0.2437 | 0.3785 | 0.2302 | 0.3511 | 0.1587 | 0.2538 |
| (6) w/o UPCF | 0.2423 | 0.3721 | 0.2426 | 0.3595 | 0.1399 | 0.2384 |
| (7) w/o TSF | 0.2445 | 0.3672 | 0.1751 | 0.2738 | 0.1293 | 0.2034 |
| (8) w/o RHF | 0.2431 | 0.3733 | 0.2376 | 0.3535 | 0.1474 | 0.2440 |
| (9) w/o CAF | 0.2452 | 0.3756 | 0.2369 | 0.3545 | 0.1529 | 0.2517 |
| H2POI | **0.2535** | **0.3830** | **0.2527** | **0.3645** | **0.1677** | **0.2597** |

**Table 4**
The experimental results of the multi-task prediction ablation study.

| $\mathcal{L}_l$ | $\mathcal{L}_g$ | $\mathcal{L}_c$ | NYC | | TKY | | CA | |
|---|---|---|---|---|---|---|---|---|
| | | | Acc@1 | MRR | Acc@1 | MRR | Acc@1 | MRR |
| ✓ | | | 0.2377 | 0.3702 | 0.2374 | 0.3532 | 0.1501 | 0.2472 |
| ✓ | ✓ | | 0.2458 | 0.3812 | 0.2439 | 0.3590 | 0.1579 | 0.2530 |
| ✓ | | ✓ | 0.2437 | 0.3795 | 0.2415 | 0.3556 | 0.1531 | 0.2507 |
| ✓ | ✓ | ✓ | **0.2535** | **0.3830** | **0.2527** | **0.3645** | **0.1677** | **0.2597** |

**Table 5**
The experimental comparison of POI, region, and category prediction performance.

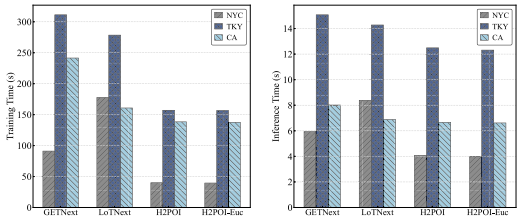
| | NYC | | TKY | | CA | |
|---|---|---|---|---|---|---|
| | Acc@1 | MRR | Acc@1 | MRR | Acc@1 | MRR |
| (1) POI | 0.2535 | 0.3830 | 0.2527 | 0.3645 | 0.1677 | 0.2597 |
| (2) Category | 0.2865 | 0.4415 | 0.3785 | 0.4865 | 0.1996 | 0.3137 |
| (3) Region | 0.9993 | 0.9993 | 0.9993 | 0.9997 | 0.9981 | 0.9988 |

### 5.5. Efficiency analysis

For **RQ4**, we conduct experiments to evaluate the efficiency of the proposed method by comparing H2POI, its Euclidean counterpart (H2POI-Euc), and two strong baselines, GETNext and LoTNext. For a fair comparison, all methods use the same batch size of 64. Fig. 7 reports the average training and inference time (in seconds) per epoch, calculated over 10 epochs. Table 6 further compares the computational efficiency in terms of trainable parameters, peak GPU memory usage, and total training time. We make the following observations. First, Fig. 7 shows that H2POI requires less average training and inference time than GETNext and LoTNext. H2POI-Euc is slightly faster than H2POI, while the difference is marginal, indicating that the additional hyperbolic operations introduce limited computational overhead. Second, as shown in Table 6, H2POI shows a clear advantage in overall training time, requiring less time than GETNext and LoTNext across all three datasets. H2POI uses fewer trainable parameters than GETNext, although it requires more parameters and GPU memory than LoTNext. Compared with H2POI-Euc, H2POI has similar parameter and memory costs. Although it requires slightly longer total training time than H2POI-Euc, it remains more training-efficient than existing methods. Overall, these results, together with the performance gains in Table 2, demonstrate that H2POI achieves superior recommendation performance with a clear advantage in training efficiency over existing methods.

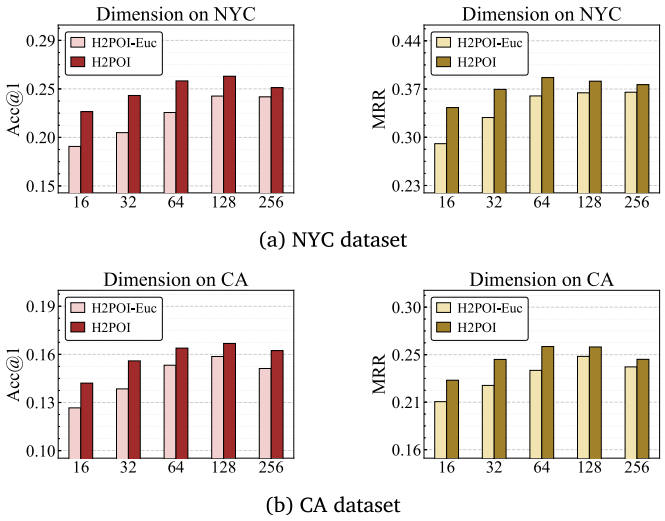
**Table 6**
The efficiency comparison in terms of the number of trainable parameters (M), peak GPU memory usage (GiB), and total training time (min).

| | Metric | GETNext | LoTNext | H2POI | H2POI-Euc |
|---|---|---|---|---|---|
| NYC | Params | 2.264 | 1.084 | 1.202 | 1.184 |
| | Memory | 2.488 | 1.666 | 3.848 | 4.342 |
| | Time | 101.94 | 172.22 | 58.13 | 52.11 |
| TKY | Params | 2.882 | 1.714 | 1.840 | 1.812 |
| | Memory | 2.081 | 0.984 | 2.604 | 2.792 |
| | Time | 306.57 | 297.54 | 204.55 | 191.00 |
| CA | Params | 3.351 | 2.179 | 2.316 | 2.281 |
| | Memory | 2.837 | 1.055 | 3.414 | 3.621 |
| | Time | 221.80 | 190.51 | 170.85 | 156.05 |




(a) Training (b) Inference

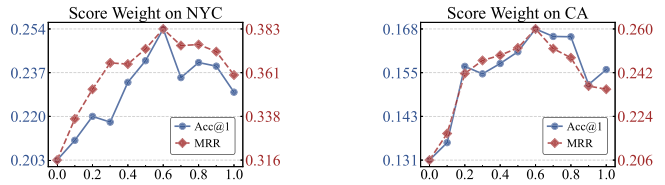
**Fig. 7.** Comparison of training and inference time.




(a) NYC dataset

(b) CA dataset

**Fig. 8.** Effect of dimension $d$ on NYC and CA datasets.




(a) NYC dataset

(b) CA dataset

**Fig. 9.** Effect of score weight $\alpha$ on NYC and CA datasets.

### 5.6. Sensitivity analysis

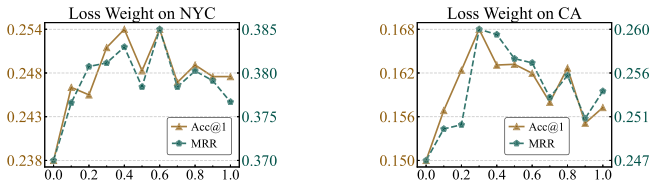
For **RQ5**, we assess H2POI's robustness through a sensitivity analysis. We study the effect of dimension, score weight, and loss weight. Results are reported on the NYC and CA datasets using Acc@1 and MRR.

#### 5.6.1. Effect of dimension

Fig. 8 shows the impact of **dimension** $d$. We compared the performance of H2POI and its Euclidean counterpart (H2POI-Euc) across different dimensions on two datasets. The results show that both methods improve as the dimension increases, suggesting that higher dimensions provide better representation capacity. However, after a certain point, the performance slightly drops, likely due to overfitting from excessive model complexity. H2POI consistently outperforms H2POI-Euc in all dimensions, particularly in lower-dimensional settings (e.g., $d \leq 64$). This demonstrates that hyperbolic space, with its exponential capacity to embed complex hierarchical structures, enables more expressive representations even with fewer dimensions.

#### 5.6.2. Effect of score weight

Fig. 9 shows the impact of **score weight** $\alpha$. As in Eq. (20), the score weight $\alpha$ controls the balance between two trajectory representation construction methods for final POI prediction. The results on both datasets show that the model achieves optimal performance when the weights of the two types of trajectory representations are balanced. When $\alpha = 0$ or $\alpha = 1$, the model relies on only one trajectory representation, effectively removing one semantic perspective from trajectory modeling, which leads to a clear performance drop. This reflects that the two proposed fusion strategies capture distinct and complementary aspects of trajectory semantics, and combining them enhances the accuracy of POI prediction. It further suggests that leveraging hyperbolic properties to construct trajectory representations with varying semantics effectively improves model performance.




(a) NYC dataset

(b) CA dataset

**Fig. 10.** Effect of loss weight $\beta$ on NYC and CA datasets.

#### 5.6.3. Effect of loss weight

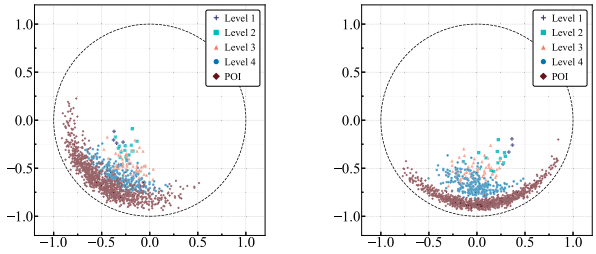
Fig. 10 shows the impact of **loss weight** $\beta$. As defined in Eq. (23), the loss weight $\beta$ is related to multi-task learning, which controls the influence of the auxiliary tasks, region and category predictions, on the overall training objective. When $\beta = 0$, the model is trained solely on the POI prediction task without multi-task learning. On both datasets, the performance drops noticeably without the auxiliary tasks, indicating that multi-task learning contributes to more robust and generalizable POI prediction. The optimal value of $\beta$ may vary across datasets, as it can be influenced by the inherent characteristics of each dataset. Overall, the results demonstrate that incorporating these semantic supervision signals ($\beta > 0$) improves POI prediction performance across both datasets, indicating that our method leverages contextual semantics to enhance the model's generalization ability.

### 5.7. Visualization

For **RQ6**, we visualize the NYC and CA datasets to intuitively explore hyperbolic space and hierarchical patterns. Fig. 11 presents the results of using CO-SNE (Guo et al., 2022) to conduct the dimensionality reduction, which maps learned Lorentz embeddings into a unit Poincaré ball with $d = 2$. Although dimensionality reduction may lead to unexpected information loss, the results still provide valuable insights. We show all region embeddings and a randomly sampled set of $1,000$ POIs to better present the results. Besides, Fig. 12 presents the distance distribution of hyperbolic embeddings from the origin for all region and POI embeddings. Prior studies (Law et al., 2019; Nickel & Kiela, 2018) have shown that the radial positions of representations in hyperbolic space can reflect hierarchical organization. As shown in Figs. 11 and 12, the learned representations exhibit clear distance patterns on both datasets. Region embeddings at different spatial levels show progressively different radial distributions, while the more fine-grained POI representations are generally located farther from the origin. This indicates that representations with different spatial granularities occupy distinct positions in hyperbolic space. Such distance patterns are consistent with the multi-level region semantics described in Section 3.2, providing representation-level evidence that H2POI effectively captures and preserves hierarchical characteristics in the learned hyperbolic space. Similar patterns are also observed between category and POI embeddings across all three datasets. Overall, the results provide complementary evidence that H2POI preserves hierarchical characteristics in the learned hyperbolic representations.

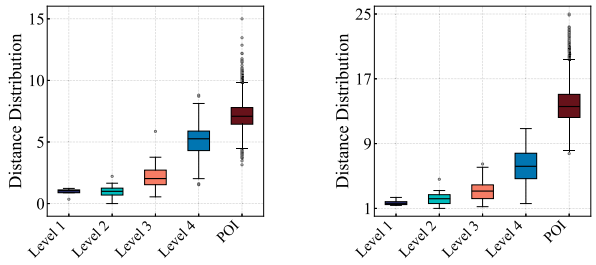
## 6. Limitations and future work

While the experimental results demonstrate H2POI's effectiveness across multiple benchmark datasets, several directions remain for further extension. First, our evaluation is based on commonly used public benchmark datasets and does not cover substantially larger POI systems, such as those containing tens of thousands of POIs. Future work may extend our hyperbolic multi-semantic modeling framework to large-scale POI recommendation with substantially larger POI sets. Second, H2POI learns user-specific representations from historical check-ins and has not explicitly considered unseen users or users with sparse histories. Future




(a) NYC dataset (b) CA dataset

**Fig. 11.** Visualization of the embeddings for regions at each level and POIs on NYC and CA datasets.




(a) NYC dataset (b) CA dataset

**Fig. 12.** Illustration of embeddings' distance distributions for regions at each level and POIs on NYC and CA datasets.

work may further explore hierarchical semantics to address these scenarios, where higher-level region and category information could provide complementary knowledge when user-POI interactions are sparse or unavailable.

## 7. Conclusion

We introduce H2POI, a novel hyperbolic next POI recommendation model that addresses the limitations of existing methods in capturing hierarchical structures and diverse semantics embedded in check-ins. Observing that hierarchical patterns are prevalent across diverse semantics, the core idea of H2POI is to systematically fuse hierarchical semantics within a unified hyperbolic framework. Experimental results show that H2POI consistently outperforms existing Euclidean and hyperbolic methods for next POI recommendation, showing that modeling these semantics in hyperbolic space improves the characterization of user mobility patterns. The ablation results confirm that modeling from both global and trajectory perspectives is an effective strategy for capturing hierarchical semantics in hyperbolic space. We further demonstrate that incorporating various semantics improves personalized POI recommendation by leveraging richer contextual information to infer user mobility intentions more robustly. Visualization results indicate that the hierarchical relationships between POIs and their attributes are preserved through hyperbolic distances. Our work highlights the potential of hyperbolic geometry as a principled foundation for next POI prediction.

## CRediT authorship contribution statement

**Hongliang Qiao:** Writing – original draft, Methodology, Formal analysis, Conceptualization; **Shanshan Feng:** Writing – review & editing, Supervision, Methodology; **Hao Wang:** Writing – review & editing; **Xutao Li:** Writing – review & editing; **Tiantian He:** Writing – review & editing; **Fan Li:** Supervision, Funding acquisition; **Puay Siew Tan:** Supervision; **Yew-Soon Ong:** Supervision, Project administration.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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